

Layer 2 Switching – Detailed Theory

1. Introduction to Layer 2 Switching

Layer 2 Switching operates at the Data Link layer (Layer 2) of the OSI model. It is primarily responsible for forwarding Ethernet frames between devices on the same local area network (LAN).

Unlike Layer 3 devices (routers), which route packets between different networks based on IP addresses, Layer 2 switches forward frames based on MAC (Media Access Control) addresses.

2. Functionality of Layer 2 Switch

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MAC Address Table (CAM Table):

A Layer 2 switch maintains a MAC address table that maps MAC addresses to switch ports. It learns the MAC addresses by inspecting the source MAC address of incoming frames and associates them with the incoming port.

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Forwarding Decision:

When a frame arrives, the switch looks at the destination MAC address:

- If the destination MAC is in the table, forward the frame out only the port associated with that MAC address (unicast).
- If the destination MAC is a broadcast address (FF:FF:FF:FF:FF:FF) or multicast, forward out all ports except the incoming port (flood).
- If the destination MAC is unknown, the switch floods the frame to all ports except the incoming port.

- **Frame Switching Methods:**

- **Store-and-Forward:** The switch receives the entire frame, checks for errors using CRC, and forwards if no errors are detected.
- **Cut-Through:** The switch starts forwarding the frame as soon as it reads the destination MAC address, which reduces latency but may forward corrupted frames.
- **Fragment-Free:** A compromise, forwarding begins after reading the first 64 bytes (the minimum Ethernet frame size), catching most errors caused by collisions.

3. Key Features of Layer 2 Switches

- **Segmentation:** Switches divide a network into multiple collision domains (one per switch port), reducing collisions and improving performance compared to hubs.
- **Broadcast Domain:** By default, all ports on a Layer 2 switch are in the same broadcast domain. Broadcast traffic is flooded to all ports.
- **MAC Address Learning and Aging:** The MAC address table entries age out after a timeout (usually 5 minutes) if no frames are received from that MAC, freeing space and allowing for dynamic network changes.

4. Switching vs. Bridging

Switching is conceptually an evolution of bridging. Both use MAC addresses to forward frames, but switches are more scalable, supporting more ports and higher performance.

5. Spanning Tree Protocol (STP)

- **Purpose:** Prevent Layer 2 loops that can cause broadcast storms and MAC table instability.
- **How:** STP creates a loop-free logical topology by blocking redundant paths while keeping a backup.
- **Standards:** IEEE 802.1D (original STP), IEEE 802.1w (Rapid STP), IEEE 802.1s (Multiple STP).

6. VLANs (Virtual LANs)

- Layer 2 switches support VLANs to segment broadcast domains logically.
- Each VLAN acts as a separate broadcast domain.
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VLAN tagging protocols: IEEE 802.1Q (most common), ISL (Cisco proprietary).

7. Trunking

- Trunks are links between switches carrying traffic for multiple VLANs simultaneously.
 - Uses tagging (802.1Q) to identify VLAN membership.
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8. Port Security

- Layer 2 switches support port security to restrict MAC addresses on ports.
- Prevents unauthorized devices from connecting.

- Can limit MAC addresses per port or specify allowed addresses.
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9. Switch Management and Monitoring

- Switches can be managed via CLI, SNMP, or web interfaces.
 - Commands like show mac address-table, show interfaces, and show spanning-tree are important for troubleshooting.
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10. Other Concepts

- **Broadcast Storms:** Excessive broadcast traffic that can cripple a network; controlled by storm control features.
- **MAC Address Flooding Attack:** A security threat where an attacker floods the switch with fake MAC addresses to overflow the MAC table, turning the switch into a hub and enabling sniffing.

